



Full Paper

OVERVIEW

01 DATASET: ML2HP

- **21 subjects** performing 17 static hand poses across 2 perpendicular LMC2 devices (Horizontal and Vertical views)
- Volume: **714,000 synchronized records**.
- **Landmark Modality Only:** Uses the 247-dimensional landmark representation (joint 3D coordinates, palm position/orientation, finger widths) without image processing.



The Leap Motion Controller 2 (LMC2) sensor

02 GEOMETRIC FEATURE ENGINEERING

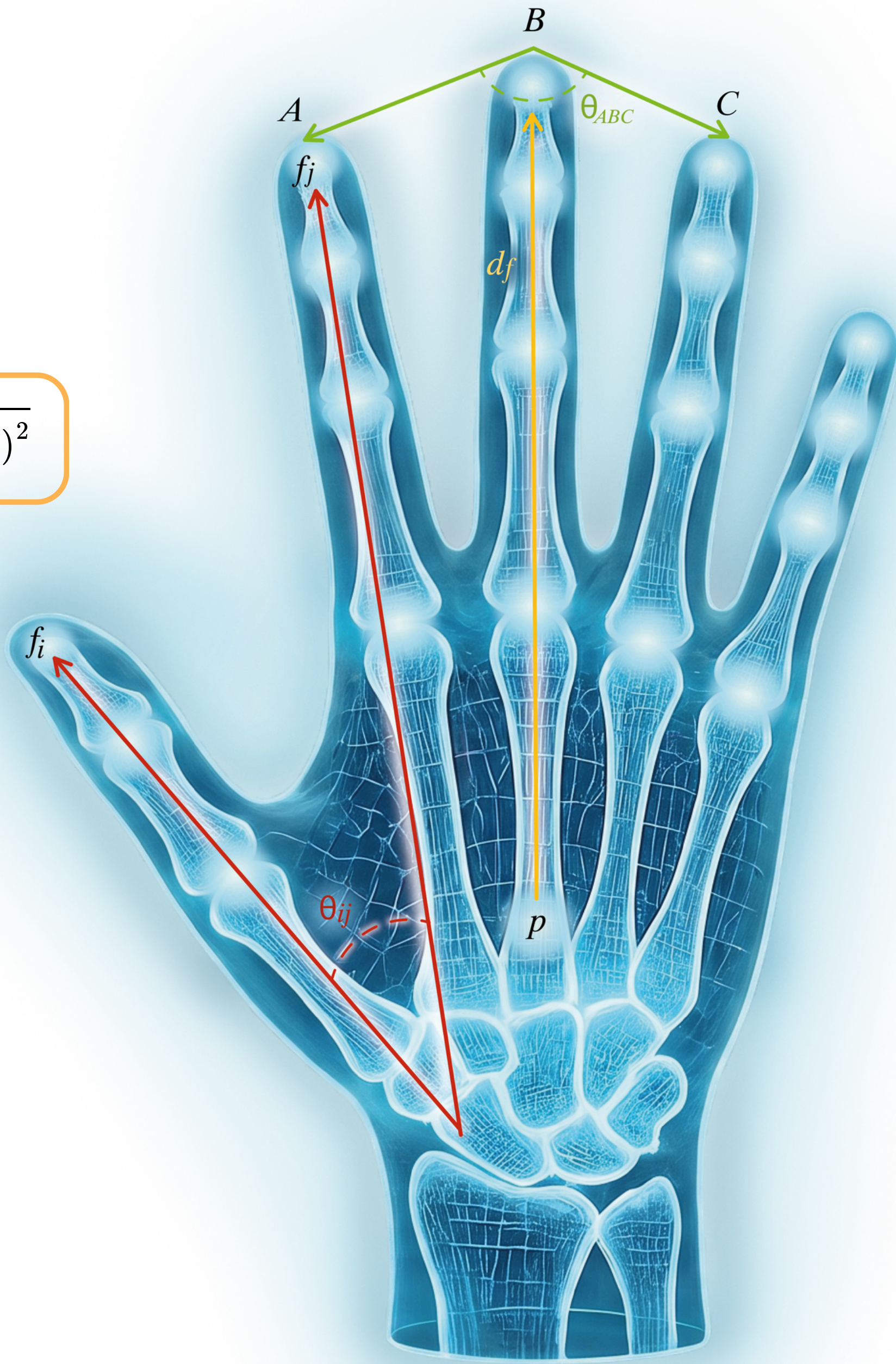
To capture intrinsic hand geometry, we enrich raw landmarks with two sets of explicit descriptors:

1. **Distance Features:** Fingertip-to-palm Euclidean distances to capture global hand proportions.
2. **Angular Features:** Palm-normalized pairwise and triplet finger angles, projected into a local palm-centered coordinate system to ensure robustness against global translation and rotation.

$$\theta_{ABC} = \arccos \left(\frac{(A - B) \cdot (C - B)}{\|A - B\| \|C - B\|} \right)$$

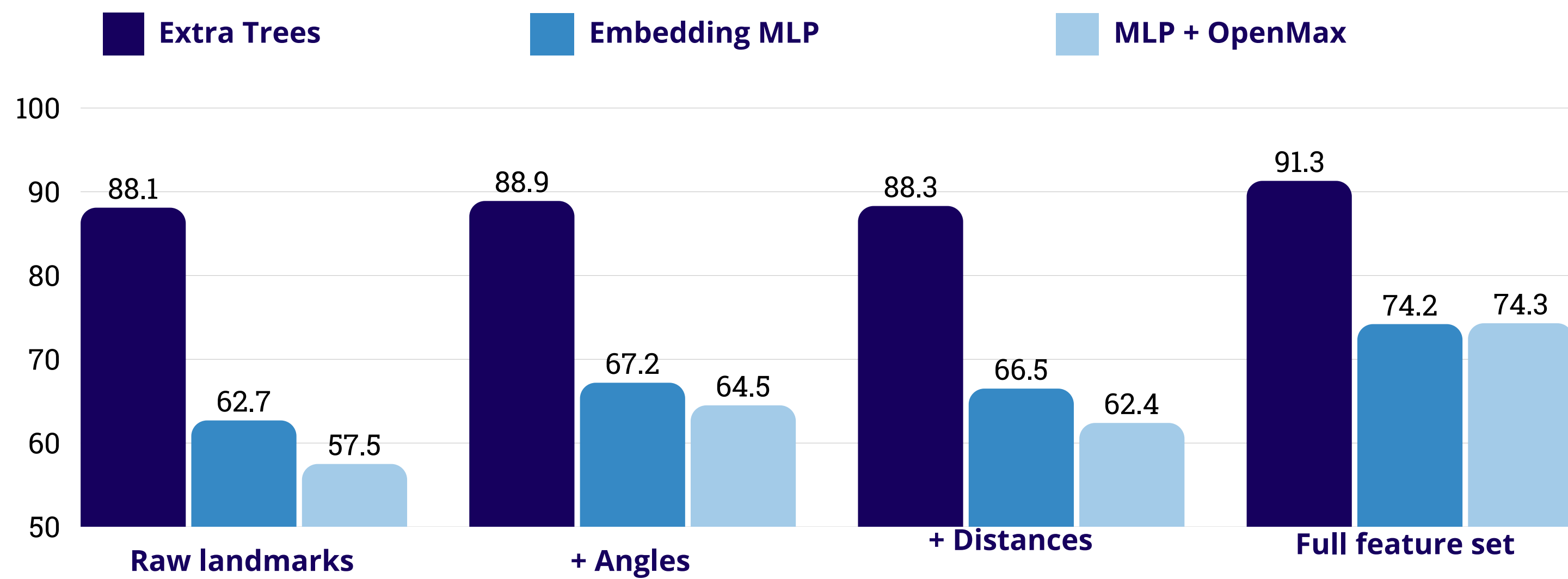
$$d_f = \sqrt{(x_f - x_p)^2 + (y_f - y_p)^2 + (z_f - z_p)^2}$$

$$\theta_{ij} = \arccos \left(\frac{\mathbf{v}'_{f_i} \cdot \mathbf{v}'_{f_j}}{\|\mathbf{v}'_{f_i}\| \|\mathbf{v}'_{f_j}\|} \right)$$



Final Representation: Concatenation of raw LMC2 attributes + 5 distances + 10 pairwise angles + triplet angles.

ABLATION

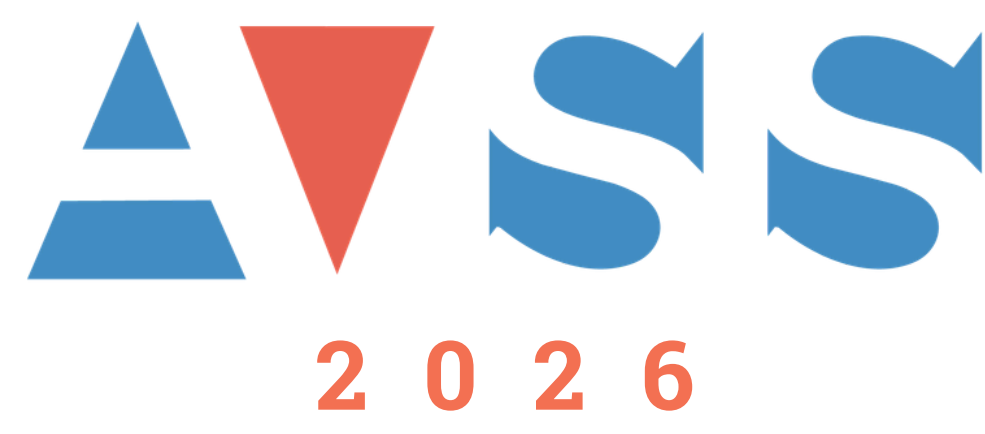


Open-set accuracy across feature configurations. Handcrafted geometric descriptors help every model; angles contribute more than distances in isolation; the neural models gain the most (+11.6 and +16.9 points) yet never overtake the ensemble.

Subject-Level Unknown-Identity Identification from Leap Motion Controller 2 Hand Landmarks

Bahar Moharrer, Susanna Cifani, Marco Raoul Marini, Luigi Cinque, Maria De Marsico

Sapienza University of Rome, Italy



03 EVALUATION PROTOCOL

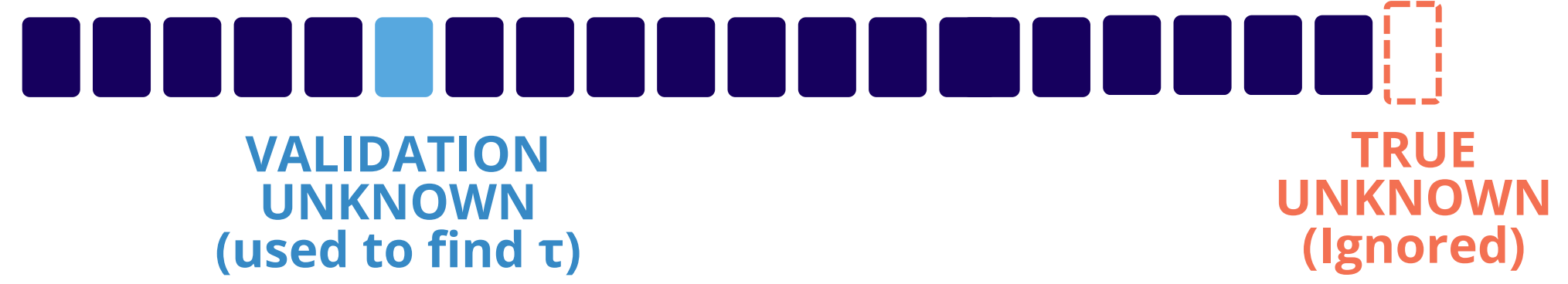
OUTER FOLD k (true unknown):

In each fold, 1 subject is held out as the **True Unknown** while the model is trained on the remaining 20 subjects. However, we must first estimate a decision threshold τ to distinguish known subjects from unknown subjects.



INNER SPLIT (threshold estimation):

To find τ , within the 20 known subjects we hold 1 out as a **Validation Unknown**. The model is trained on the training split of the remaining 19 subjects, then evaluated on the validation split of all 20 subjects to compute Unknown Scores.



Using these Unknown Scores, we compute the FAR and FRR curves across candidate thresholds to select τ at the EER point (FAR = FRR).

Model	Unknown Score (s_{unk})
Extra Trees	$1 - p_{\max}$
Embedding MLP + Centroids	$1 - \max_k \cos(z, c_k)$
MLP + OpenMax	$p_{\text{OpenMax}}(\text{unk} \mid x)$

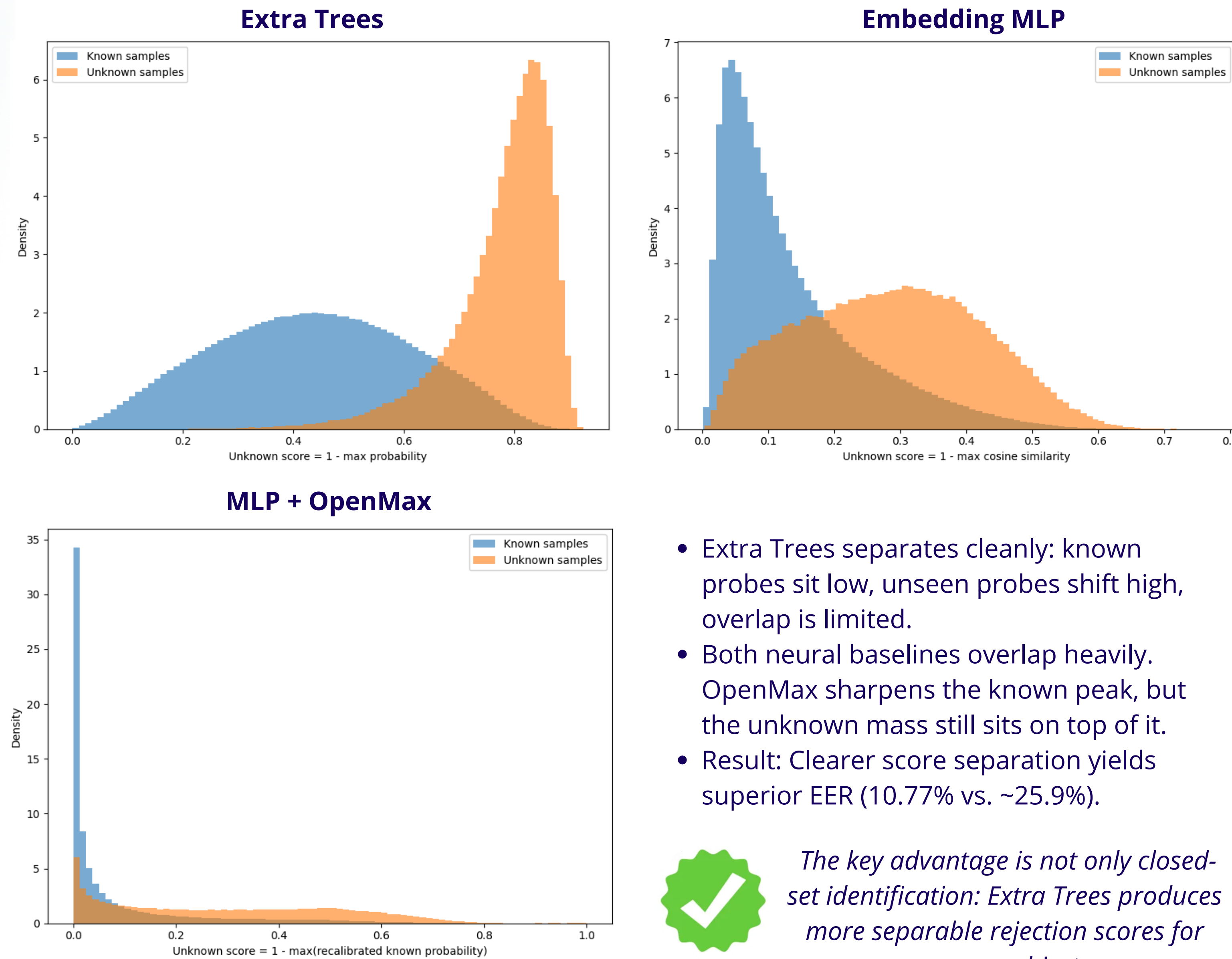
A higher score indicates stronger evidence for rejecting the probe as an unknown identity

Having obtained the threshold τ , we can now return to the Outer Fold k: the model can finally be trained on all 20 enrolled subjects and evaluated across all 21 subjects using the fixed τ .



Zero Data Leakage: The threshold τ was tuned without ever seeing the True Test Unknown!

04 UNKNOWN SCORE DISTRIBUTIONS



MAIN RESULTS: 21 LOSO FOLDS, MEAN $\pm$ STD

Model	Closed-set Accuracy	Open-set Accuracy	EER
Extra Trees	$99.49 \pm 0.06\%$	$91.29 \pm 1.50\%$	$10.77 \pm 2.84\%$
Embedding MLP + Centroids	$94.78 \pm 0.25\%$	$74.23 \pm 1.66\%$	$25.88 \pm 3.27\%$
MLP + OpenMax	$95.10 \pm 0.28\%$	$74.32 \pm 1.62\%$	$25.95 \pm 3.74\%$

While all models excel at recognizing enrolled users (Closed-set), Extra Trees halves the EER, proving its superiority in handling unseen unknowns.